Games and Information

Midterm Examination — Solutions and Grading Rubric

Prof. M. Brandt · Spring 2023 · 60 points

General grading policy. Award full credit for any correct method. An arithmetic slip that is carried through consistently loses at most 1 point per part. A final answer with no supporting work earns at most half the points of that part. Graders should write the rubric line numbers used next to each deduction.

Problem 1 (15 points)

(a) Players: firm 1 and firm 2. Firm 1 has a single type ($c_1 = 30$); firm 2's type set is $\{L, H\}$ with $c_2^L = 18$, $c_2^H = 36$. Action sets: $q_i \in [0, \infty)$. Beliefs: firm 1 assigns probability $1/3$ to L and $2/3$ to H (common prior); firm 2 knows both costs. A pure strategy for firm 1 is a single quantity q_1; a pure strategy for firm 2 is a *type-contingent plan* (q_2^L, q_2^H). Payoffs are profits, and firm 1 maximizes expected profit.

Rubric (3 pts).

- 1 pt: players, type sets and prior stated correctly.
- 1 pt: firm 2's strategy is a function of its type (the key idea; no credit if firm 2's strategy is a single number).
- 1 pt: action sets and (expected) profit payoffs.

(b) Type L maximizes $(120 - q_1 - q_2 - 18)q_2$, so $q_2^L(q_1) = \dfrac{102 - q_1}{2}$. Type H maximizes $(120 - q_1 - q_2 - 36)q_2$, so $q_2^H(q_1) = \dfrac{84 - q_1}{2}$.

Firm 1 maximizes $\frac{1}{3}(90 - q_1 - q_2^L)q_1 + \frac{2}{3}(90 - q_1 - q_2^H)q_1 = (90 - q_1 - \bar{q}_2)q_1$, where $\bar{q}_2 = \frac{1}{3}q_2^L + \frac{2}{3}q_2^H$. Hence $q_1(\bar{q}_2) = \dfrac{90 - \bar{q}_2}{2}$: firm 1 best-responds to the *expected* output of firm 2.

Rubric (4 pts).

- 1 pt each for the two type-specific best responses (first-order conditions).
- 2 pts for firm 1's best response to the expected quantity. Deduct 1 pt if firm 1 is (incorrectly) given two type-dependent quantities.

(c) From (b), $\bar{q}_2 = \frac{1}{3} \cdot \frac{102 - q_1}{2} + \frac{2}{3} \cdot \frac{84 - q_1}{2} = 45 - \frac{q_1}{2}$. Substituting, $q_1 = \frac{1}{2}\big(90 - 45 + \frac{q_1}{2}\big)$, so $\frac{3}{2}q_1 = 45$ and

$$q_1^* = 30, \qquad q_2^{L*} = 36, \qquad q_2^{H*} = 27.$$

Prices: $P = 120 - 30 - 36 = 54$ when $c_2 = 18$ and $P = 120 - 30 - 27 = 63$ when $c_2 = 36$. (For reference: profits are 1296 for type L, 729 for type H, and $\frac{1}{3}(24 \cdot 30) + \frac{2}{3}(33 \cdot 30) = 900$ expected for firm 1.)

Rubric (4 pts).

- 2 pts: correct system and $q_1^* = 30$.
- 1 pt: both type quantities 36 and 27.
- 1 pt: both prices.

- A common error is to replace firm 2 by an "average" firm with cost 30 and report $q_1 = q_2 = 30$: award 1 pt out of 4.

(d) With complete information, Cournot with costs (c_1, c_2) gives $q_1 = \frac{120-2c_1+c_2}{3}$ and $q_2 = \frac{120-2c_2+c_1}{3}$.

- $c_2 = 18$: $q_1 = 26$, $q_2 = 38$ (price 56).
- $c_2 = 36$: $q_1 = 32$, $q_2 = 26$ (price 62).

The high-cost type produces 27 under incomplete information versus 26 under complete information. Under incomplete information firm 1 produces 30, less than the 32 it would produce against a known high-cost rival; quantities are strategic substitutes, so the high-cost type responds to the lower q_1 by producing more. (Symmetrically, the low-cost type produces less: $36 < 38$.)

Firm 1's expected profit with the information is $\frac{1}{3}(26)^2 + \frac{2}{3}(32)^2 = \frac{676+2048}{3} = 908$, versus 900 in the BNE. The information is worth $908 - 900 = 8$ to firm 1.

Rubric (4 pts).

- 2 pts: the two complete-information equilibria (1 pt each).
- 1 pt: correct comparison ($27 > 26$) with the strategic-substitutes explanation; comparison without explanation earns 0.5.
- 1 pt: value of information = 8. Students who compute 908 but forget to subtract earn 0.5.

Problem 2 (15 points)

(a) $\mathbb{E}[u_R(S)] = 6\mu$ and $\mathbb{E}[u_R(J)] = 3$. So S is a best response iff $\mu \geq 1/2$ (with indifference at $\mu = 1/2$).

Rubric (2 pts).

- 1 pt: expected payoffs; 1 pt: threshold 1/2.

(b) *Candidate 1: $T \to P$, $A \to N$.* Bayes' rule gives $\mu(P) = 1$ and $\mu(N) = 0$, so the employer plays S after P and J after N. Type T gets $12 - 3 = 9$; deviating to N gives $5 < 9$. Type A gets 5; deviating to P gives $12 - 8 = 4 < 5$. Hence this is a PBE:

$$\text{Sender: } (T : P,\ A : N); \quad \text{Employer: } S \text{ after } P,\ J \text{ after } N; \quad \mu(P) = 1,\ \mu(N) = 0.$$

Candidate 2: $T \to N$, $A \to P$. Then $\mu(N) = 1$, $\mu(P) = 0$, so S after N and J after P. Type A gets $5 - 8 = -3$ but would get 12 by switching to N. Not a PBE.

So there is exactly one separating PBE.

Rubric (5 pts).

- 1 pt: beliefs from Bayes' rule in candidate 1.
- 1 pt: employer's responses.
- 2 pts: both incentive constraints checked ($9 \geq 5$ and $5 \geq 4$), 1 pt each.
- 1 pt: candidate 2 ruled out with a specific profitable deviation.

(c) *Pooling on N.* On path $\mu(N) = 2/5 < 1/2$, so the employer plays J after N and both types get 5. Type A's best payoff from P is $4 < 5$, so A never deviates. Type T would get $9 > 5$ from P if the employer answered S, so the employer must answer P with J, which is optimal iff the off-path belief satisfies $\mu(P) \leq 1/2$. The PBE is:

$$\text{both types } N; \quad J \text{ after } N,\ J \text{ after } P; \quad \mu(N) = 2/5,\ \mu(P) \in [0, 1/2].$$

Pooling on P. On path $\mu(P) = 2/5$, so the employer plays J after P; type A gets $5 - 8 = -3$, while N yields at least 5 whatever the employer does. Not a PBE for any off-path belief.

Rubric (4 pts).

- 1 pt: on-path belief $2/5$ and response J.
- 2 pts: correct off-path belief condition $\mu(P) \leq 1/2$ (1 pt if the student only says "pessimistic beliefs" without the threshold).
- 1 pt: pooling on P ruled out.

(d) No. In the pooling PBE type A earns 5, and the most it could ever earn from P is $12-8 = 4 < 5$, so P is equilibrium-dominated for A. Type T earns 5 and could earn 9 from P if hired as Senior. The Intuitive Criterion therefore requires $\mu(P) = 1$; the employer then plays S after P and type T deviates. The pooling PBE fails the Intuitive Criterion (the separating PBE has no off-path messages and passes trivially).

Rubric (2 pts).

- 1 pt: identify that P is equilibrium-dominated for A but not for T.
- 1 pt: conclude $\mu(P) = 1$ breaks the equilibrium.

(e) With $c_A = 6$: in the candidate $T \to P$, $A \to N$, type A gets 5 but mimicking gives $12 - 6 = 6 > 5$. In the candidate $T \to N$, $A \to P$, type A gets $5 - 6 = -1$ but N gives 12. Both fail, so no separating PBE exists. (Separation requires $12 - c_A \leq 5$, i.e., $c_A \geq 7$.)

Rubric (2 pts).

- 1 pt for each separating candidate ruled out.

Problem 3 (15 points)

(a) In a second-price auction a bidder's bid affects only *whether* she wins, not what she pays (she pays the other bid x). Bidding v wins exactly when $x < v$, i.e., exactly when winning is profitable. Bidding $b > v$ changes the outcome only when $v < x < b$, where she now wins and pays $x > v$ (a loss). Bidding $b < v$ changes the outcome only when $b < x < v$, where she now loses a profitable win. So bidding v is weakly dominant.

Revenue is the lower of the two values: $\mathbb{E}[\min\{v_1, v_2\}] = 2 + \frac{1}{3} \cdot 4 = \frac{10}{3}$, i.e., about \$333.33.

Rubric (3 pts).

- 2 pts: dominance argument covering both over- and under-bidding (1 pt if only one direction).
- 1 pt: revenue $10/3$.

(b) Suppose the rival bids $\alpha+\beta v'$ with $\beta > 0$. A bid b wins iff $v' < (b-\alpha)/\beta$, which has probability $\big(\frac{b-\alpha}{\beta} - 2\big)/4$ when this lies in $[0,1]$. The bidder maximizes

$$(v-b)\cdot\frac{1}{4}\Big(\frac{b-\alpha}{\beta} - 2\Big).$$

The first-order condition is $-\big(\frac{b-\alpha}{\beta} - 2\big) + \frac{v-b}{\beta} = 0$, i.e., $b = \frac{v+\alpha+2\beta}{2}$. For this to equal $\alpha + \beta v$ for all v we need $\beta = \frac{1}{2}$ and $\alpha = \frac{\alpha+2\beta}{2}$, so $\alpha = 2\beta = 1$. Hence $b(v) = 1 + v/2$.

Verification: rival bids lie in $[2,4]$. For $b \in [2,4]$, $\Pr(\text{win}) = \frac{b-2}{2}$ and the payoff $(v-b)\frac{b-2}{2}$ is strictly concave with maximizer $b = \frac{v+2}{2} \in [2,4]$. A bid below 2 never wins, and a bid above 4 wins with probability one but pays more than bidding 4. So $b(v) = 1 + v/2$ is a best response for every $v \in [2,6]$.

Rubric (5 pts).

- 1 pt: correct winning probability.
- 2 pts: first-order condition and matching coefficients.
- 2 pts: global verification (concavity, plus bids outside $[2,4]$). 1 pt if only the FOC is checked.
- Students who incorrectly use the $U[0,1]$ formula $b = v/2$ earn at most 1 pt.

(c) The winner is the bidder with the higher value, so revenue is $\mathbb{E}\big[1 + \frac{1}{2}\max\{v_1, v_2\}\big] = 1 + \frac{1}{2}\big(2 + \frac{2}{3}\cdot 4\big) = 1 + \frac{7}{3} = \frac{10}{3}$.

Rubric (3 pts).

- 2 pts: revenue is the expected bid of the highest type; 1 pt: correct value.

(d) A bidder with $v = 5$ wins with probability $\Pr(v' < 5) = \frac{3}{4}$ in both formats.

- SPA: she pays v' when she wins; $\mathbb{E}[v' \mid v' < 5] = 3.5$, so the expected payment is $\frac{3}{4}\cdot 3.5 = \frac{21}{8} = 2.625$.
- FPA: she pays her bid $b(5) = 3.5$ when she wins, so the expected payment is again $\frac{3}{4}\cdot 3.5 = \frac{21}{8}$.

This illustrates the *revenue equivalence theorem*: with independent private values and risk-neutral bidders, two auctions with the same allocation rule (the highest value wins) and the same payoff for the lowest type (here $v = 2$ earns zero) give every type the same expected payment, hence the seller the same expected revenue.

Rubric (4 pts).

- 1 pt: winning probability 3/4.
- 1 pt each: the two expected payments.
- 1 pt: names revenue equivalence and at least one of its conditions (same allocation / same lowest-type payoff / IPV / risk neutrality).

Problem 4 (15 points)

(a) A seller offers her car iff $p \geq \theta$, so the cars on the market are uniform on $[0,p]$. Buyers' expected value is $\mathbb{E}[\theta \mid \theta \leq p] + 3 = p/2 + 3$. Setting $p = p/2 + 3$ gives $p = 6$. Sellers with $\theta \leq 6$ sell: half of all cars.

Rubric (4 pts).

- 1 pt: sellers with $\theta \le p$ sell.
- 2 pts: equilibrium condition $p = p/2 + 3$ and $p = 6$.
- 1 pt: fraction $1/2$.

(b) Every trade creates surplus 3. The cars with $\theta \in (6, 12]$ (probability $1/2$) are not traded, so the expected loss is $\frac{1}{2} \cdot 3 = 1.5$ per car in the population (\$1,500).

Rubric (3 pts).

- 2 pts: identify the untraded cars and the per-trade gain of 3; 1 pt: 1.5.

(c) At price $p > 0$ the cars offered are uniform on $[0, p]$, worth $\mathbb{E}[1.5\,\theta \mid \theta \le p] = 0.75\,p < p$ to buyers. So buyers are unwilling to pay any positive price, and the only equilibrium price is $p = 0$, at which only $\theta = 0$ is offered: essentially no trade, even though buyers value every car 50% more than its owner.

Rubric (3 pts).

- 2 pts: $0.75p < p$ for all $p > 0$; 1 pt: conclusion of complete unraveling.

(d) Certifying yields $\theta + 3 - 1 = \theta + 2 > \theta$, so no seller keeps her car if she can certify. A seller certifies iff $\theta + 2 \ge p_u$, i.e., $\theta \ge p_u - 2$; sellers with $\theta < p_u - 2$ sell uncertified (and $p_u > \theta$ for them). Buyers' valuation of uncertified cars gives

$$p_u = \mathbb{E}[\theta \mid \theta < p_u - 2] + 3 = \frac{p_u - 2}{2} + 3 \quad \Longrightarrow \quad p_u = 4.$$

So sellers with $\theta < 2$ sell uncertified at 4, sellers with $\theta \ge 2$ certify and receive $\theta + 3$ (net $\theta + 2$), and *all* cars are traded. The allocation is efficient, but certification fees of $1 \times \Pr(\theta \ge 2) = 5/6$ per car are spent, so the net loss falls from 1.5 to $5/6$.

Rubric (5 pts).

- 1 pt: certifying beats keeping for every seller.
- 1 pt: cutoff rule $\theta \ge p_u - 2$.
- 2 pts: fixed-point equation and $p_u = 4$ (cutoff $\theta = 2$).
- 1 pt: all cars traded (the comparison of losses is not required for full credit).